



STRATHMORE INSTITUTE OF MATHEMATICAL SCIENCES
MASTER OF SCIENCE IN DATA SCIENCE & ANALYTICS
ASSIGNMENT 1-
DSA 8505: Bayesian Statistics

DATE: 11th Mar 2025

Instruction

- (a) Use python in this Assignment
- (b) Provide your commented scripts
- (c) Scan and submit your answer through the Google Classroom by 23h59, 11th Mar 2025.

You are provided with a dataset named marketing.csv.

This dataset contains information on marketing campaigns of a Portuguese banking institution and includes details such as customer demographics, contact data, and campaign outcomes. It is suitable for analyzing the effectiveness of marketing strategies and their impact on customer engagement.

Your task is to apply Bayesian hypothesis testing to analyze the effectiveness of a marketing campaign using real-world data. You will use Python to perform statistical analysis based on the Bayesian framework discussed in the lecture notes.

1. Load the dataset into Python, give an overview of the important variables for this task and perform data cleaning (such as handling missing values, encoding categorical variables, etc.).
2. State the Bayesian Hypotheses
3. Perform Bayesian Analysis
 - Use prior probabilities based on past marketing campaigns.
 - Compute likelihoods using data from the current campaign.
 - Use Bayes' theorem to compute the posterior probabilities of H_0 and H_1 .
 - Calculate and interpret the Bayes Factor (BF).

4. Based on the Bayes Factor, determine whether there is evidence supporting the effectiveness of the marketing campaign.
5. Plot the prior, likelihood, and posterior distributions.
6. Provide a well-documented report with your findings.